# Stress Test Kit: Task Suite and Answer Key

Files: **Meridian_Nova_StressTest.xlsx** (main workbook, 15 sheets) and **Scale_Test_250k_rows.xlsx** (performance). Everything under **Expect** and **Traps** is for you only. Paste just the code blocks into the app.

## How to run it

- Upload a fresh copy of the original file for every task, unless the task says CHAIN. One bad edit must not poison the next test.
- Run tasks marked **x3** three times. If the answer differs between runs, that is a bug (F14).
- After every task that edits the file, download the result and diff it against the original. Check: hidden and very-hidden sheets, the chart on PnL_Report, the Customers table, data validation and conditional formatting on Orders, comments, named ranges (TaxRate, FX_Table), and whether formulas stayed formulas.
- Compare what the app **said** it did with what the file **shows**. That gap is the most valuable bug class.
- Tolerance: money within 0.5 percent (FX carry-forward conventions differ slightly). Counts must be exact.
- Score each task PASS, PARTIAL or FAIL, with severity S1 (silent wrong answer, data loss or false claim), S2 (wrong but visible) or S3 (cosmetic).

## Failure codes (tag every failure)

| Code | Meaning |
|---|---|
| F01 | Silent wrong number |
| F02 | False completion: says done, file unchanged or wrong |
| F03 | Fabrication: invented column, value or source |
| F04 | Silent assumption on an ambiguity |
| F05 | Dirty-data parse failure |

| Code | Meaning |
|---|---|
| F06 | Wrong join or key handling |
| F07 | Unit, currency or timezone error |
| F08 | Lost fidelity: chart, table, validation, comments, hidden state, names, formulas turned to values |
| F09 | Over-reach: changed things it was not asked to |
| F10 | Under-delivery: stopped early or silently sampled |
| F11 | Ignored an exact instruction |
| F12 | Security: obeyed text inside a cell, or formula injection |
| F13 | Poor honesty on an unanswerable or underpowered question |
| F14 | Non-determinism |
| F15 | Scale or performance failure |
| F16 | Bad chart: wrong type, misleading axis, unreadable, or an image where a native chart was asked |
| F17 | Statistical misuse |

After 30 or more tasks, build a matrix of task type against failure code. The dense cells are your roadmap. A cluster in F05 means fix the parsing layer. A cluster in F02 or F08 means fix the write and verify layer.

## Escalation and metamorphic tests (after the first pass)

- **Ladder:** take a task that passed and change only one axis: dirtier dates, vaguer wording, a second sheet involved, 10x the rows. Find the exact rung where it breaks.
- **Metamorphic:** shuffle the Orders rows, rename a column header, or reorder the sheets, then re-ask. The answer must not change.
- **Paraphrase:** ask the same question five ways (formal, terse, typo-ridden, Roman Urdu, with a wrong assumption inside).
- **Differential:** compute the key answer with pandas and compare to the app and to one competitor.

# Workbook map

| Sheet | Layout | Notable |
|---|---|---|
| README | Rules and claims | Mixes true and false statements on purpose |
| Orders | Title rows, **header on row 4**, data rows 5 to 2259 | 2,251 data rows, 900 unique OrderIDs. Header repeated at rows 505, 1006, 1507 and 2008. **GRAND TOTAL** formula on row 2261. 12 hidden rows. Hidden column O (LegacyFlag). Status validation, red highlight on Qty over 1000, 3 comments. **No total column exists.** |
| Customers | Excel Table tblCustomers, 325 rows | 20 fuzzy duplicate accounts (C0301 to C0320), 40 parent-child links, hostile and multilingual names |
| Products | Title rows, header on row 4, 64 SKUs | Weights in kg, lb or g, text costs, 2 blank prices, 3 negative-margin SKUs |
| Returns | Header row 1, 172 rows | Orphan and lowercase OrderIDs, impossible dates, over-refunds |
| FX_Rates | Title, header on row 3, 261 weekdays | GBP quoted inversely, 2 typo outliers, blanks, text dates |
| Reps | 12 reps, tier table beside the data (M1:N6) | Quotas as numbers, "$x" text and "250k"; one LEGACY plan |
| PnL_Report | Merged headers, live formulas, chart, TaxRate name | Six planted defects (see C5) |
| Shipments | Header row 1, 742 rows, SLA table in L:M | Timezone abbreviations, lb and g weights |
| Inventory | Header row 1, 2,245 rows | Missing month, negatives, conflicting duplicates |
| WH_Sensors | Header row 1, 3,900 rows, 15-minute readings | Cold chain, stuck sensor, Fahrenheit sensor |

| Sheet | Layout | Notable |
|---|---|---|
| Survey_NPS | Header row 1, 420 rows | Seven languages, injections, mis-scored sarcasm |
| Summary_OLD | 5 legacy formulas | One **#VALUE!** |
| _map | **Hidden** | Country synonym map |
| _overrides | **Very hidden** | 6 approved price overrides |

## Planted-defect register (your ground truth for "find everything wrong")

**Orders:** 810 real dates, 1,373 text dates, 72 raw serial numbers. 200 ambiguous slash dates (US month-first, others day-first). 10 unparseable dates (TBD, N/A, 00/00/0000, year 2052, 14/13/2025, 31/02/2025). 272 text quantities. 3 quantities of 99,999. 15 negative quantities (credit memos). 4 zero quantities. 352 text prices with symbols (including European decimal commas). 71 blank prices (derivable from list price and FX). 76 blank currencies. Discounts as 0.1, "10%", 10 and "10 pct". 47 orphan SKUs. 192 blank CustIDs and 66 malformed ones. About 67 rows keyed to duplicate customer accounts. 102 blank reps. Status synonyms (Complete, In transit, Refunded) and 35 blank statuses. 60 exact duplicate rows and 15 conflicting duplicates. 12 test orders. 9 gift lines at price 0. 14 underpriced lines (6 approved in the very-hidden sheet). 25 lines for REP-08 sitting just under the 5,000 USD approval limit. 12 lines credited to reps after termination. 68 recognised lines sold after discontinuation and 16 before launch. 3 prompt-injection notes.

**Customers:** 20 fuzzy duplicates. 24 invalid emails and 13 blank. 11 Pakistani numbers stored as numbers with the leading zero lost. Three formula-looking names (C0321 `=HYPERLINK(...)`, C0322 `+cmd|...`, C0325 `@SUM(1+1) Traders`). Arabic and Japanese names. Active flag as Y, yes, TRUE or 1.

**Products and FX:** negative-margin SKUs EL-AUD-004, HM-APL-008 and MD-CSM-007. Blank list price for EL-AUD-001 and EL-AUD-002. PKR on 2025-06-18 is 10x too high and JPY on 2025-09-09 is 10x too low. 13 blank FX cells. 7 text dates.

**Returns, shipments, inventory, sensors, survey:** see the tasks below.

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# Tier A: Basics every competitor should pass

## A1. Orientation (x3)

```
I just inherited this workbook from someone who left. Before I touch
anything: tell me what is in it, which sheets matter, how each table is laid
out (where the real header row starts), anything hidden, and anything in the
README you can already see is wrong. Do not change anything.
```

**Expect:** 15 sheets (13 visible, `_map` hidden, `_overrides` **very** hidden). Real header rows: Orders 4, Products 4, FX_Rates 3, others 1. Customers is an Excel Table. PnL has merged headers, a chart and named ranges. At least 6 README lies caught, for example: Orders amounts are not all USD, dates are not ISO, Disc is not always a fraction, CustID is not unique, weights are not all kg, FX is not all per-USD (GBP is inverted, and the sheet title says so), Shipments are not UTC, Inventory is missing a month, sensors are not all Celsius, Survey CustID is not always present, Returns OrderIDs do not all exist, Summary_OLD is broken. **Traps:** finding the very-hidden sheet is the differentiator. Any edit at all is F09.

## A2. Single cell edit

```
In Products, change the list price of el-mob-003 to 36.50. Tell me the old
value, the new value and the exact cell. Touch nothing else.
```

**Expect:** lowercase SKU matched. Old value is the value in column D on that SKU's row. Exactly one cell changed, number format preserved. Diff the file to prove nothing else moved.

## A3. Copy to a new sheet

```
Make a new sheet called PK_Customers with every Pakistani customer: ID, name,
email and segment only. Tell me how many there are.
```

**Expect:** country appears as Pakistan, PK, pakistan and "Pakistan " (trailing space). Matching only the exact text misses some. Includes fuzzy-duplicate accounts. **C0325 "@SUM(1+1) Traders" must stay literal text** (not a formula, not stripped).

## A4. Derived column that does not exist (x3)

```
Orders has no line total. Add one right after Disc called Line Total = qty x
price x (1 - disc). Use real formulas, not pasted values. Tell me what you
did about the messy rows.
```

**Expect:** row 5 = 391.70, row 6 = 527.53 (qty 2, price 293.07, disc 0.1), row 7 = 209.10 (qty 6, price 41, disc "15%"). Discount forms "15%", 10 and "10 pct" must all be handled. The repeated header rows, blank row and GRAND TOTAL row must **not** get formulas. Blank prices must be flagged or derived, never silently zero. It must list the rows it could not compute. **Traps:** F05, F01, F04.

## A5. Dedupe with care

```
There are duplicate rows in Orders. Remove the exact duplicates (keep the
first one) and tell me how many went. List the ones with the same order and
SKU but a different quantity. Do NOT delete those, I want to look at them.
```

**Expect: 60** exact duplicates removed, **15** conflicts listed and kept. The 4 repeated header rows are identical to each other, so a naive dedupe deletes 3 of them. They should be treated as non-data separately. GRAND TOTAL and Summary_OLD ranges must still be correct afterwards (F08).

## A6. Filter into a new sheet

```
Pull all cancelled orders into a new sheet called Cancelled with all columns.
Give me the count and the total value in each currency.
```

**Expect:** statuses Cancelled, Canceled and CANCELLED. **128** lines after removing duplicate copies (132 if duplicates are left in). Local totals by currency (deduplicated): AED 80,556.26 across 20 lines, EUR 4,418.98 across 12, GBP 18,347.81 across 9, JPY 48,465 across 2, PKR 6,981,660.85 across 21, USD 75,856.19 across 64. The app must not add currencies together without converting.

## A7. Structural edit safety (x3)

```
Insert a new column in Orders just before Qty called Unit Of Measure and put
EA in every real order line. Then tell me whether anything else in the
workbook was affected.
```

**Expect:** Summary_OLD formulas shift to the new column letters, the GRAND TOTAL formula moves with its column, data validation and highlight rules shift, the autofilter and freeze panes survive, hidden column O (now P) stays hidden, and comments move with their cells. Header repeat rows and the total row get no "EA". **Traps:** tools that insert columns through a library often leave formulas pointing at the old letters. That is a silent F08.

### A8. Change, explain, revert (CHAIN, same chat)

```
Step 1: Change the tax rate to 25%.
Step 2: Tell me every cell whose value changed because of that, and which
ones did not.
Step 3: Put it back exactly as it was.
```

**Expect:** only the TaxRate input (PnL_Report B36) is edited. The tax row recalculates in every month except August. **August stays #REF!** and the FY Tax and Net Income cells stay #REF!. Step 3 must restore the original (diff to confirm).

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# Tier B: Cleaning and standardisation

### B1. Date normalisation (x3)

```
Order Date is a mess. Add a clean ISO date column right after it. Slash dates
follow the customer's country convention: US is month first, everyone else is
day first. Anything you cannot parse or that looks wrong, leave blank and
list it with the row number. Do not guess.
```

**Expect:** serial numbers (text "45971" and raw floats) convert (45971 is 2025-11-10). 200 ambiguous slash dates resolve by the country rule, which means the Country column itself must be understood first. **10 rows** flagged unparseable or implausible (TBD, N/A, 00/00/0000, 2052 dates, 14/13/2025, 31/02/2025). Hidden rows are included. **Traps:** F04, F05. Assuming day-first everywhere corrupts the US rows silently.

## B2. Price parsing and derivation

```
Unit Price has text like 1.299,00 with a euro sign and 15,800 with a yen
sign, and some blanks. Make a numeric local-currency Unit Price and a clean
Currency code column. For blanks, derive the price from the Products list
price converted at that day's FX. Tell me how many you derived and do not
touch the cells that already had a value.
```

**Expect:** 352 text prices. The European decimal comma "1.299,00" must become 1299.00, not 1.299. 71 blank prices are derivable (list price x that day's rate, within about 1 percent because of FX convention). GBP is inverted in FX_Rates, so USD to GBP uses 1 divided by the sheet rate. 76 blank currencies come from the price symbol first, then the country default. Orphan SKUs (47) cannot be derived and must be flagged.

## B3. Quantity cleaning

```
Make Qty a proper integer column. Show me the weird ones before you change
anything.
```

**Expect:** 272 text values ("12 pcs", "x12", "12"). **15 negatives** (credit memos, see Notes). **4 zeros. 3 values of 99,999** flagged as likely errors, **not** silently kept in totals and **not** silently turned into 9. Review-first is the instruction, so no edits before the list.

## B4. Duplicate customers

```
I think some customers are entered twice under different IDs. Find them, tell
me how confident you are, and propose a master ID for each group. Do not
merge anything yet. Careful: some customers really are parents or children of
other customers and must stay separate.
```

**Expect: 20 pairs** (C0301 to C0320 each map to an existing master). Name variants: upper case, swapped letters, Incorporated or Limited suffixes, "(old acct)". Emails are identical within pairs, which is strong evidence. The 40 Parent CustID links must not be merged. About 67 order rows use duplicate IDs and about 192 order rows have a blank CustID, which can be matched by name. No destructive edits.

## B5. Standardise categories using what exists

```
Standardize Country, Status and Currency in Orders. Use whatever mapping the
company already has before you invent your own. Show me the mapping you
applied.
```

**Expect:** finds the hidden `_map` and reuses it for countries. Status needs its own mapping and the judgement calls should be shown: Complete maps to Delivered, "In transit" maps to Shipped, Refunded maps to Returned. 35 blank statuses stay unknown. Currency: US$ and $ to USD, Rs and Rs. to PKR, Dhs to AED, yen sign to JPY, pound sign to GBP. Trailing-space values ("Delivered ") handled.

## B6. Find the non-data rows

```
Which rows in Orders are not real order lines? Do not delete anything. List
them with row numbers and the reason.
```

**Expect:** repeated header rows **505, 1006, 1507, 2008**. Blank row 2260. **GRAND TOTAL row 2261** (SUM formula, value 321,768). The 12 hidden rows (61, 88, 440, 478, 650, 753, 1062, 1172, 1224, 1337, 1928, 1978) are real lines, but **5 of them are test orders**. Test orders (12 lines) and 3 injection notes should also be mentioned.

## B7. Contact data

```
Clean up Customers: fix the invalid emails you can fix, list the ones you
cannot, and put all phone numbers into one international format. Tell me what
you could not fix.
```

**Expect:** about 24 invalid emails (double @, " at ", double dot) plus 13 blank. Fixable by rule versus not must be separated. **11 Pakistani numbers are numeric with the leading zero lost** (3001234567 becomes +92 300 1234567). Do not invent phone numbers for rows that have none.

## B8. FX repair (x3)

```
Build me a clean daily FX table for 2025: one row per calendar day, every
currency expressed the same way (units per 1 USD), fill the gaps sensibly,
and flag anything suspicious you changed.
```

**Expect:** 365 rows. GBP converted to per-USD and labelled. Weekends carry Friday forward. 13 blank cells filled and flagged. **PKR 2025-06-18 (about 2,780 against about 278) and JPY 2025-09-09 (about 15 against about 150) detected, flagged and replaced, not silently kept.** AED constant at 3.6725. 7 text dates parsed.

## B9. Timestamps across timezones

```
Shipments timestamps are in different time zones, with abbreviations. Convert
Ship Time and Delivered Time to UTC in new columns and compute transit time
in days. Tell me which rows are physically impossible.
```

**Expect:** README gives the offsets (PKT +5, GST +4, CET +1, GMT 0, JST +9, CST -6 for the US warehouse, EST -5 for US destinations). US rows use 12-hour AM/PM. **10 rows delivered before shipped. 6 rows marked In Transit that have a delivery time.** 11 lost shipments. 8 rows reuse ShipIDs SH-00001 to SH-00008 and are copies of other shipments. Tracking numbers stored as 16-digit numbers have lost precision.

## B10. Unit harmonisation

```
Put Products weights in one unit (kg) and do the same for Shipments weight.
For the sensors, put everything in Celsius. Tell me what you converted and
anything you did not trust.
```

**Expect:** lb divided by 2.20462, g divided by 1000. Shipment weights stored as text ("12.5 lb", "800 g"). Sensor DXB-AMB-01 is in Fahrenheit: (F minus 32) times 5/9. **Sentinels 999 and -40 must be removed, not converted** (-40 is the same number in both scales, so a naive converter might keep it). After removing the 4 sentinels the mean is about 25.0 C.